

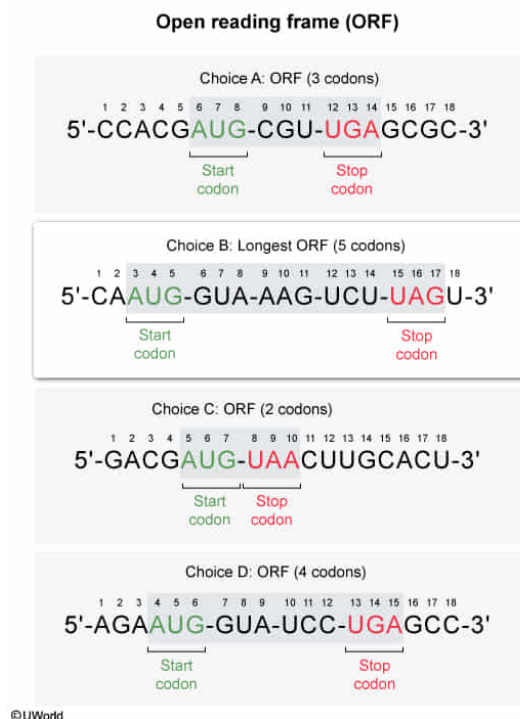
Which of the following mRNA strands contains the longest open reading frame?

- ☐ A. 5'-CCACGAUGCGUUGAGCGC-3' [15%]
- ✓ ☒ B. 5'-CAAUGGUAAAGUCUUAGU-3' [54%]
- ☐ C. 5'-GACGAUGUAACUUGCACU-3' [6%]
- ☐ D. 5'-AGAAUGGUAUCCUGAGCC-3' [23%]

Correct

53% answered correctly

Explanation:



The **genetic code** consists of sequences of three consecutive nucleotides in DNA or RNA called **codons**, each of which codes for a specific amino acid or a stop signal during translation. A **reading frame** is a set of consecutive codons that do not overlap. Any given strand of mRNA has three **possible reading frames**, depending on which nucleotide is the first in a codon.

An **open reading frame (ORF)** is a portion of a reading frame that can be translated by ribosomes because it contains both a start codon (AUG) and a stop codon (UAA, UAG, or UGA) within the **same reading frame**.

After **transcription** of DNA to RNA, the resulting mRNA strand is translated in a 5' to 3' direction by the ribosome to generate a polypeptide. Translation of the mRNA sequence begins with a **start codon**, which codes for the amino acid methionine. **Translation continues** until the ribosome **reaches a stop codon** in the same reading frame as the start codon.

Based on the ORF definition, the mRNA sequence in **Choice B** has a *start codon* (AUG) that begins at the *third nucleotide* and a stop codon (UAG) that begins at the *fifteenth nucleotide*. There are three additional codons between the start and stop codon of this sequence, making the ORF of this sequence 5 codons long, which is **longer than any other ORF shown**. (Note that this strand also contains UAA, but this sequence is **out of frame**, so it is not translated as a

stop codon.)

(Choice A) This mRNA sequence contains an ORF that is only 3 codons long.

(Choice C) The start codon in this sequence is immediately followed by a stop codon, making the ORF only 2 codons long.

(Choice D) This mRNA sequence contains an ORF that is only 4 codons long.

Educational objective:

An open reading frame (ORF) is a set of codons within a strand of mRNA that can be translated by a ribosome. An ORF begins with a start codon (AUG) and ends with a stop codon (UAA, UAG, or UGA). The start and stop codons must be in the same reading frame.

Time spent:0 sec

Subject:
Biology

QID:402520

System:
Molecular Biology